

Hashir Bin Zahid

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SUMMARY

Mathematically rigorous and performance-focused graduate student in Information Systems (SAP major, Cybersecurity minor). Strong foundation in C++ systems programming, numerical optimization, and AI-driven modeling. Experienced in backtesting predictive models, time series forecasting, and applying linear algebra, probability theory, and Monte Carlo methods to real-world systems. Proven ability to engineer scalable, low-latency applications, from medical management platforms to e-commerce CMS tools, and to deliver high-visibility projects such as the Half-Life: Alyx FakeVR mod with 500K+ downloads. Adept at blending technical depth with creative problem-solving and community engagement. Thrives in high-pressure environments where mathematical rigor and technical depth drive innovation and measurable impact.

EDUCATION

Central Michigan University, Mt Pleasant, MI	Expected May 2027
<i>Master of Science in Information Systems, SAP Major</i>	4.0 GPA
University of Houston, Houston, TX	May 2025
<i>Bachelor of Science in Computer Science, Data Science Minor</i>	3.6 GPA

WORK EXPERIENCE

Infodat International Inc., Houston, TX	June 2024 - May 2025
<i>Full-Stack Developer</i>	
- Developed full-stack applications using Spring Boot, Java, Angular, and Docker - Created and maintained RESTful APIs that improved data exchange efficiency across microservices - Optimized microservices for latency reduction, implemented reproducible workflows for audit compliance	
Data Annotations, Remote	May 2024 - March 2025
<i>AI Specialist</i>	
- Labeled and curated large-scale datasets for natural language, improving training accuracy and reducing noise - Validated reinforcement learning models with statistical rigor, curated datasets for quantitative fairness metrics	

TECHNICAL PROJECTS

Half-Life: Alyx FakeVR Mod
- Created a mod that enables full keyboard and mouse gameplay for Half-Life: Alyx, bypassing VR hardware requirements - Implemented core mechanics including hand interactions and weapon handling to replicate immersive gameplay without VR - Garnered half a million downloads and international media coverage on the project
E-Medic - Health Management System
- Engineered a medical management system in C# with patient onboarding, appointment scheduling, and payment tracking - Integrated Syncfusion PDF generation to automate medical report creation - Designed robust admin panels with full CRUD functionality for managing patients, lab results, doctors, and appointments
Portal Gun - Unreal Engine 4
- Developed the Portal Gun mechanic in Unreal Engine 4, replicating core teleportation physics and portal rendering - Applied linear algebra, vector transformations, and spatial matrices to design seamless transitions between portals with accurate spatial logic and collision handling.
Half-Life: Alyx MK Edition - Unreal Engine 4 Recreation
- Recreated a playable portion of Half-Life: Alyx in Unreal Engine 4, enabling mouse and keyboard control without VR - Implemented custom assets, scripted interactions, and environmental triggers to mirror original gameplay mechanics
E-Commerce CMS - C++ Management System
- Built a comprehensive e-commerce content management system in C++ with separate portals for customers and employees - Implemented BinaryTree class in C++ with optimized traversal and insertion, leveraging memory management and algorithmic efficiency for latency-sensitive applications.

RESEARCH PROJECTS

Price Prediction Using AI
- Developed & compared 12 distinct prediction models using metrics such as MSE and directional accuracy - Trained a custom stacked model that surpassed all other models in price prediction. - Developed and backtested stacked predictive models for time series forecasting, achieving superior accuracy (MSE=7) with rigorous statistical validation.
Malware Evasion Techniques
- Researched adversarial strategies and obfuscation, applying probabilistic reasoning and system-level analysis - Explored techniques including droppers, time bombs, in-memory execution, and command-line manipulation to disable antivirus or create exclusions

CERTIFICATIONS

Celonis Rising Star Program Celonis	Dec. 2025
Nvidia Fundamentals of Accelerated Data Science Nvidia	Aug. 2024
Energy Leadership University of Houston	April 2024
Azure Fundamentals uCertify	May 2023

VOLUNTEER EXPERIENCE

Business Communication RSO, Mt. Pleasant <i>Event Manager</i>	Oct. 2025 - Current
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- Coordinate logistics for campus-wide networking events, including venue booking, vendor outreach, and promotional strategy to maximize student engagement
- Design and implement feedback systems to evaluate event impact, leading to a 30% improvement in attendance

Central Michigan University Student Pantry, Mt. Pleasant <i>Volunteer Assistant</i>	Aug. 2025 - Current
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- Assist students in need by distributing essential items including food, hygiene products, and daily supplies
- Maintain organized inventory and restock shelves to ensure consistent availability of high-demand items

AWARDS AND HONORS

University of Houston

- Deans Distinguished Scholars List, 2023
- Graduated with Honors, Cum Laude, 2025

Lone Star College

- Presidential List, 2021-2023

Entrepreneurship Fair

Most Innovative Award - November 2025

SKILLS

Programming & Development

- C++, C#, HTML/CSS, Java, JavaScript, Python,
- Spring Boot, Angular, Node.js
- RESTful APIs, MySQL, MongoDB, Microsoft Access
- Docker, Git, GitHub, Visual Studio, Eclipse
- CMS development, inventory and logistics systems

AI & Data Science

- Model training and evaluation (MSE, directional accuracy)
- Stacked model design and performance tuning
- Data labeling, prompt engineering, feedback iteration

Creative & Communication

- YouTube content creation (567K+ views)
- Community engagement through modding and tutorials